

An Internship Report
Of
Mastercard Cybersecurity Job Simulation

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 25SCS1003000702

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Batch: 2025-2029

CANDIDATE'S DECLARATION

I, MD AYAN ALAM, do hereby declare that the internship report titled “Mastercard Cybersecurity Job Simulation” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: MD AYAN ALAM

Roll No.: 25SCS1003000702

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Mastercard and Forage for providing me the opportunity and the structured platform to gain practical, hands-on exposure to the work of a cybersecurity and security awareness team. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: MD AYAN ALAM

Roll No.: 25SCS1003000702

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3. INTERNSHIP COMPLETION CERTIFICATE



Md Ayan Alam
Cybersecurity Job Simulation

Certificate of Completion
September 20th, 2026

Over the period of September 2026, Md Ayan Alam has completed practical tasks in:
Design a phishing email simulation
Interpret phishing simulation results



Tom Brunskill
Co-Founder of Forage

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Cybersecurity Job Simulation offered by Mastercard on the Forage virtual work experience platform. The simulation was completed in September 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The programme is designed to let students immerse themselves in the type of work that employees of Mastercard's Security Awareness Team carry out each day.

The primary aim of this internship was to build a practical understanding of phishing – one of the most common and damaging forms of social-engineering attack – by designing a phishing e-mail simulation for an organisation and then interpreting the results of that simulation to recommend appropriate security awareness training. On successful completion of all the practical tasks, a Certificate of Completion was issued by Forage on behalf of Mastercard.

4.2 Organization Profile

Mastercard is a global technology company in the payments industry, connecting consumers, financial institutions, merchants and governments across more than 200 countries and territories. Securing the data and transactions that pass through its network is central to its operations, and its Security Awareness Team works to reduce human-factor risk across the organisation through training, simulation and awareness campaigns.

Forage is an online platform that partners with leading global employers to create free, open-access virtual job simulations. These simulations are built from the real work carried out inside those organisations and are designed to help students build experience, grow confidence and demonstrate motivation before applying for roles.

4.3 Problem Statement

Phishing continues to be one of the leading causes of data breaches, because it targets people rather than technology. Even organisations with strong technical controls remain exposed when employees cannot recognise a fraudulent message. Students of computer science typically learn about social engineering only at a definitional level, with little practice in constructing a realistic simulated attack, measuring how an organisation

responds to it, or converting those measurements into a targeted training response. The problem addressed by this internship was to practically learn and apply the security awareness workflow – from designing a phishing simulation, to interpreting its results, to recommending the training needed to close the gaps it reveals.

4.4 Project Objectives

- To understand the role and day-to-day responsibilities of a security awareness function within a large organisation.
- To study the common indicators of a phishing e-mail, including sender spoofing, urgency, pretexting and malicious links.
- To design a realistic phishing e-mail simulation suitable for testing employee awareness.
- To interpret the results of a phishing simulation and identify the departments and behaviours that carry the highest risk.
- To recommend appropriate, targeted security awareness training based on the findings of the simulation.
- To successfully complete all practical tasks and obtain the Certificate of Completion.

4.5 Scope of the Project

The scope of this internship covers the security awareness cycle as practised within an enterprise environment – beginning with an understanding of phishing and social-engineering techniques, moving to the design of a simulated phishing campaign, and concluding with the analysis of simulation results and the recommendation of remedial training. The internship was limited to a virtual, simulation-based learning environment provided by Forge on behalf of Mastercard, and did not involve access to, or deployment on, any live organisational production system or real employee data.

4.6 Technologies and Tools Used

- Security Domain: Phishing and social engineering, security awareness training, human-factor risk
- Simulation Design: Phishing e-mail construction, pretexting and lure design, sender and link spoofing indicators

- Analysis: Interpretation of phishing simulation results, click-rate and report-rate based risk assessment
- Documentation: Preparation of written deliverables and recommendations for a security awareness team
- Learning Platform: Forge virtual job simulation platform, Mastercard task briefs and model answers

4.7 System Architecture

The general workflow followed during the simulation for a security awareness programme can be summarized in the following layers:

- Awareness Layer: The organisation's security awareness programme defines the objectives, the target audience and the acceptable risk thresholds.
- Design Layer: A simulated phishing e-mail is constructed with a realistic pretext, sender identity and call to action, so that it reflects the attacks employees actually receive.
- Delivery Layer: The simulated message is delivered to selected groups of employees under controlled conditions, with no real payload or data collection beyond the simulation itself.
- Measurement Layer: Employee responses – such as messages opened, links clicked and phishing attempts reported – are captured as simulation results.
- Analysis Layer: The results are interpreted to identify the departments, roles and behaviours that represent the highest human-factor risk.
- Training Layer: Targeted security awareness training is recommended and delivered to the groups identified, closing the loop back to the awareness layer.

4.8 Methodology

The internship followed a structured, task-wise methodology in which each task presented a real workplace scenario, followed by a written deliverable that was compared against a model answer prepared by the Mastercard team, as summarized in the table below.

Task	Module	Description
Task 1	Design a Phishing Email Simulation	Studied Mastercard's security awareness programme, reviewed real phishing indicators, and designed a simulated phishing e-mail to test employee awareness.
Task 2	Interpret Phishing Simulation Results	Analysed the reported results of the phishing simulation, identified the departments and behaviours at highest risk, and recommended targeted security awareness training.

Table 4.1: Task-wise methodology followed during the simulation

Note: There was no stipend associated with this internship.

Each task concluded with a written submission and a comparison against the sample response provided by Mastercard, allowing the approach taken to be evaluated against professional practice. On completion of both tasks, the Certificate of Completion was issued through Forage.

4.9 Expected Outcomes

- Ability to explain the role of a security awareness function and the human-factor risks it is designed to reduce.
- Practical understanding of phishing and social-engineering techniques, and of the indicators that identify a fraudulent message.
- Capability to design a realistic phishing e-mail simulation for an organisation.
- Skill to interpret simulation results and identify the groups and behaviours that carry the greatest risk.
- Ability to recommend targeted, evidence-based security awareness training.
- Successful completion of both practical tasks, resulting in the award of the Mastercard Cybersecurity Job Simulation Certificate of Completion through Forage.

4.10 Certificates of Completion and Communication Proof

The Certificate of Completion issued by Forage on behalf of Mastercard, confirming successful completion of the practical tasks of the Cybersecurity Job Simulation, is attached in Chapter 3 of this report. The official welcome and completion e-mails received from Forage are attached below as further communication proof.

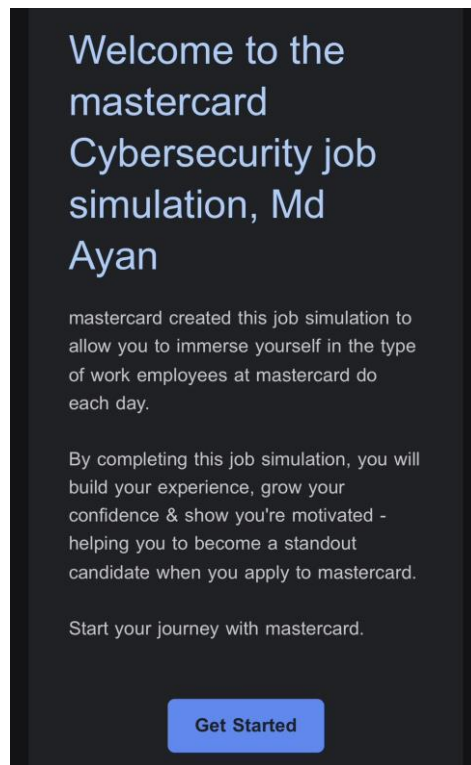


Figure 4.1: Welcome e-mail for the Mastercard Cybersecurity Job Simulation

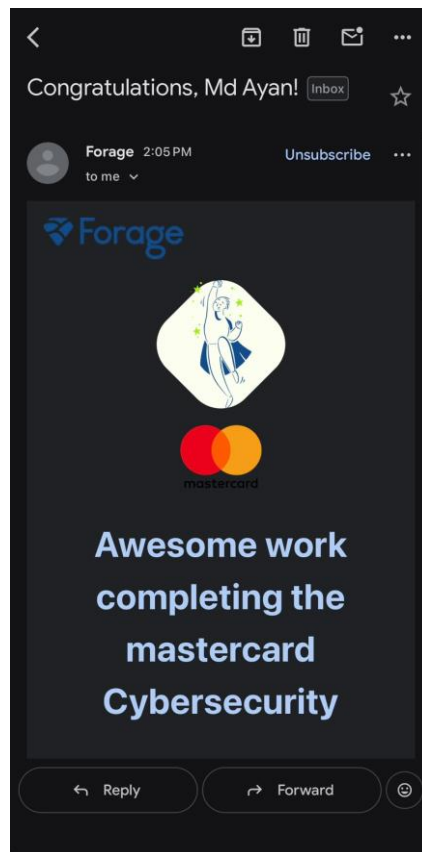


Figure 4.2: Completion confirmation e-mail received from Forage

5. BIBLIOGRAPHY/REFERENCES

1. Mastercard, “Cybersecurity Job Simulation,” Forage, Certificate of Completion, September 2026.
2. Forage, Virtual Work Experience Programs, <https://www.theforage.com/>
3. Mastercard, Security and Trust, <https://www.mastercard.com/>
4. National Institute of Standards and Technology (NIST), “Phishing,” Computer Security Resource Center, <https://csrc.nist.gov/>
5. Cybersecurity and Infrastructure Security Agency (CISA), “Avoiding Social Engineering and Phishing Attacks,” <https://www.cisa.gov/>